MILITARY SPECIFICATION

CRYSTAL UNITS, QUARTZ, PRECISION,

GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-C-49468, dated 14 September 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 21

4.9.5 (d), delete in its entirety.

PAGE 22

4.9.7.1, delete and substitute:

4.9.7.1 "Capacitance, shunt (see 3.11.1). Crystal units shall be tested in accordance with method 305 of MIL-STD-202. Since the crystal unit is actually a three terminal device and a capacitor is a two terminal device, the value measured for the shunt capacitance depends on the electrical connection of the third (case) terminal. The electrical connection to the case or to the metal shield (see 4.9.1.1) shall be as follows:

a. Two-terminal or coaxial measurement system (one-port): The case shall be connected to the instrument ground. A measurement shall be taken between each terminal and ground, the average of the measurements shall be the reported value.

b. Three-terminal (I-V) measurement system: The case shall be connected to one of the crystal unit terminals. A measurement shall be taken between the terminals and the measurement shall be recorded. The case shall then be connected to the other terminal and the measurement taken and recorded. The two measurements shall be averaged and the results recorded as the test result.

c. Two-port transmission measurement system: The case shall be connected to the instrument ground and the shunt capacitance shall be computed from

$$C_{shunt} = C_o + (1/2)(C_{S1} + C_{S2})$$

where C_{S1} and C_{S2} are the input and output capacitances and C_o is the capacitance from active terminal to active terminal see (6.5.10.1)."

4.9.7.2.1, the equations for L_1 and C_1, delete and substitute:

$$L_1 = [1/(8\pi\Delta f)]\ [X_1(\Delta f) - X_1(-\Delta f)]$$

$$C_1 = \frac{1}{(2\pi f_r)^2 L_1}$$

4.9.8, the equation for the quality factor, Q, delete and substitute:

$$Q = \frac{1}{2\pi f_r R_1 C_1}$$

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:10Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 17, 99
DLA - ES

User activities:
Navy - CG, MC, AS
Air Force - 19

Preparing activity
Army - ER

Agent:
DLA - ES

(Project 5955-0671)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:10Z
Check the source to verify that this is the current version before use.